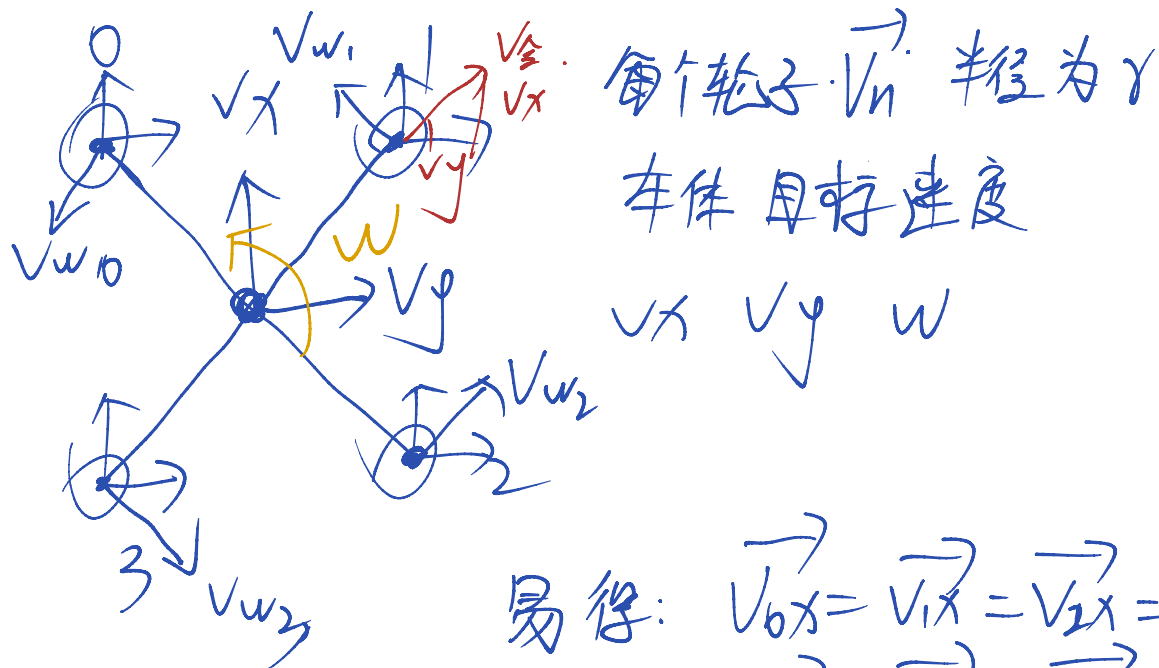


NOTEBOOK

舵轮运动学





易得: $\vec{V}_{0x} = \vec{V}_{1x} = \vec{V}_{2x} = \vec{V}_{3x} = V_x$
 $\vec{V}_{0y} = \vec{V}_{1y} = \vec{V}_{2y} = \vec{V}_{3y} = V_y$

$$|wr| = |V_{w0}| = |V_{w3}| = |V_{w2}| = |V_{w1}|$$

规定 $\rightarrow$ 为 x 正方向 用 i 表示.

$\rightarrow$ 为 y 正方向 用 j 表示.

$$\vec{V}_0 = (V_x + V_{w0} \cos 135^\circ, V_y + V_{w0} \cos 135^\circ)$$

$$\vec{V}_1 = (V_x + V_{w1} \cos 45^\circ, V_y + V_{w1} \cos 135^\circ)$$

$$\vec{V}_2 = (V_x + V_{w2} \cos 45^\circ, V_y + V_{w2} \cos 45^\circ)$$

$$\vec{V}_3 = (V_x + V_{w3} \cos 135^\circ, V_y + V_{w3} \cos 45^\circ)$$

$$\Downarrow V_{w0}, V_{w1}, V_{w2}, V_{w3} \text{ 共 } 4 \text{ 个}$$

$$\vec{V}_0 = (v_x - \omega r \frac{r_2}{2}, v_y - \omega r \frac{r_2}{2})$$

$$\vec{V}_1 = (v_x + \frac{r_2}{2} \omega r, v_y - \omega r \frac{r_2}{2})$$

$$\vec{V}_2 = (v_x + \frac{r_2}{2} \omega r, v_y + \omega r \frac{r_2}{2})$$

$$\vec{V}_3 = (v_x - \frac{r_2}{2} \omega r, v_y + \omega r \frac{r_2}{2})$$

对于舰艏 仅须速度的模长和大小。

速度大小

$$|\vec{V}_0| = \sqrt{v_{x0}^2 + v_{y0}^2}$$

$$\therefore |\vec{V}_0| = \sqrt{(v_x - \frac{r_2}{2} \omega r)^2 + (v_y - \omega r \frac{r_2}{2})^2}$$

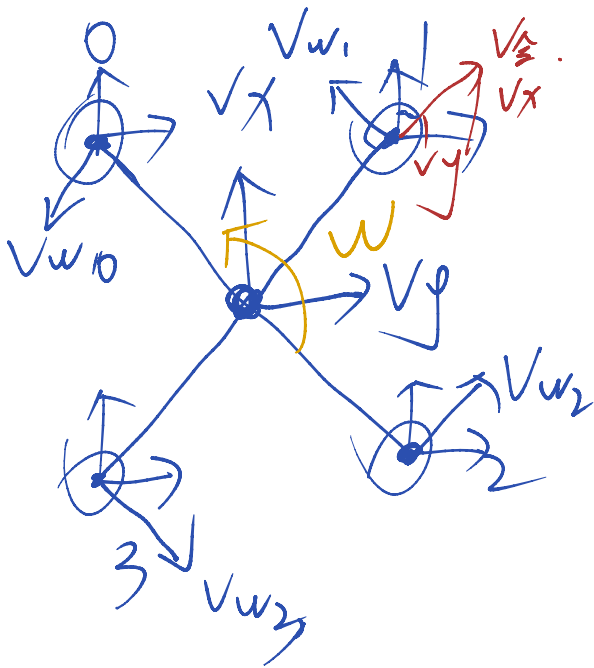
$$|\vec{V}_1| = \sqrt{(v_x + \frac{r_2}{2} \omega r)^2 + (v_y - \omega r \frac{r_2}{2})^2}$$

$$|\vec{V}_2| = \sqrt{(v_x + \frac{r_2}{2} \omega r)^2 + (v_y + \omega r \frac{r_2}{2})^2}$$

$$|\vec{V}_3| = \sqrt{(v_x - \frac{r_2}{2} \omega r)^2 + (v_y + \omega r \frac{r_2}{2})^2}$$

角度大小

设 θ 为与 $\vec{V}_0$ v_x 方向的夹角



$$\theta_0 = \tan^{-1} \left(\frac{v_y - w r \frac{r_2}{2}}{v_x - \frac{r_2}{2} w r} \right)$$

$$\theta_1 = \tan^{-1} \left(\frac{v_y - w r \frac{r_2}{2}}{v_x + \frac{r_2}{2} w r} \right)$$

$$\theta_2 = \tan^{-1} \left(\frac{v_y + w r \frac{r_2}{2}}{v_x + \frac{r_2}{2} w r} \right)$$

$$\theta_3 = \tan^{-1} \left(\frac{v_y + w r \frac{r_2}{2}}{v_x - \frac{r_2}{2} w r} \right)$$

角度

```
void chassis_vector_to_M6020_wheel_angle(fp32 vx_set, fp32 vy_set, fp32 wz_set, fp32 wheel_angle[4], int8_t need_flip[4])
{
    if(!(vx_set == 0 && vy_set == 0 && wz_set == 0)) // 防止除数为零
    {
        wheel_angle[0] = atan2((vy_set - wz_set * MOTOR_DISTANCE_TO_CENTER * SQRT2), (vx_set - wz_set * MOTOR_DISTANCE_TO_CENTER * SQRT2));
        wheel_angle[1] = atan2((vy_set - wz_set * MOTOR_DISTANCE_TO_CENTER * SQRT2), (vx_set + wz_set * MOTOR_DISTANCE_TO_CENTER * SQRT2));
        wheel_angle[2] = atan2((vy_set + wz_set * MOTOR_DISTANCE_TO_CENTER * SQRT2), (vx_set + wz_set * MOTOR_DISTANCE_TO_CENTER * SQRT2));
        wheel_angle[3] = atan2((vy_set + wz_set * MOTOR_DISTANCE_TO_CENTER * SQRT2), (vx_set - wz_set * MOTOR_DISTANCE_TO_CENTER * SQRT2));
    }
}
```

速度

```
static void chassis_vector_to_M3508_wheel_speed(fp32 vx_set, fp32 vy_set, fp32 wz_set, fp32 wheel_speed[4], int8_t r
{
    wheel_speed[0] = sqrt( pow(vy_set - wz_set * MOTOR_DISTANCE_TO_CENTER * SQRT2, 2)
    + pow(vx_set - wz_set * MOTOR_DISTANCE_TO_CENTER * SQRT2, 2)
    );
    wheel_speed[1] = sqrt( pow(vy_set - wz_set * MOTOR_DISTANCE_TO_CENTER * SQRT2, 2)
    + pow(vx_set + wz_set * MOTOR_DISTANCE_TO_CENTER * SQRT2, 2)
    );
    wheel_speed[2] = sqrt( pow(vy_set + wz_set * MOTOR_DISTANCE_TO_CENTER * SQRT2, 2)
    + pow(vx_set + wz_set * MOTOR_DISTANCE_TO_CENTER * SQRT2, 2)
    );
    wheel_speed[3] = sqrt( pow(vy_set + wz_set * MOTOR_DISTANCE_TO_CENTER * SQRT2, 2)
    + pow(vx_set - wz_set * MOTOR_DISTANCE_TO_CENTER * SQRT2, 2)
    );
}
```